

Question 1 — Student Performance Prediction (Classification)

A university wants to predict whether a student will **Pass or Fail** based on:

- Study Hours
- Attendance (%)
- Assignment Score
- Previous GPA

Tasks

1. Create a dataset manually (at least 20 records)
2. Split data into training and testing sets (80-20 split)
3. Apply feature scaling
4. Train a **Logistic Regression** model
5. Make predictions on test data
6. Evaluate using:
 - Accuracy
 - Confusion Matrix
 - Classification Report

Expected Outcome

Predict whether a **new student will Pass or Fail**

Question 2 — Email Spam Detection (Classification)

A company wants to detect whether an email is **Spam or Not Spam**.

Dataset features:

- Email Length
- Number of Links
- Number of Special Characters
- Contains "Free" (Yes/No)

Tasks

1. Create dataset manually (minimum 20 rows)
2. Split dataset into train and test
3. Train **K-Nearest Neighbors (KNN)** classifier
4. Make predictions
5. Compare results with **Logistic Regression**
6. Print accuracy of both models

Expected Outcome

Compare which model performs better:

- Logistic Regression
 - KNN
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Question 3 — House Price Prediction (Regression)

A real estate company wants to predict **house price** based on:

- House Size (sq ft)
- Number of Rooms
- Location Score
- Age of House

Tasks

1. Create dataset (20-30 records)
2. Split data (80-20)
3. Apply feature scaling
4. Train **Linear Regression** model
5. Make predictions
6. Evaluate using:
 - MSE
 - RMSE
 - R^2 Score

Question 4 — Employee Salary Prediction (Regression)

A company wants to predict employee salary based on:

- Years of Experience
- Education Level
- Skill Score
- Age

Tasks

1. Create dataset (minimum 25 records)
2. Split dataset
3. Train **Linear Regression**
4. Train **Decision Tree Regressor**
5. Compare R^2 score of both models
6. Identify best model

Question 5 — Customer Purchase Prediction (Classification)

An online store wants to predict whether a customer will **Buy or Not Buy**.

Features:

- Age
- Income
- Previous Purchases
- Time Spent on Website

Tasks

1. Create dataset manually
2. Split data
3. Apply feature scaling
4. Train:
 - a. Logistic Regression
 - b. KNN
5. Compare accuracy
6. Predict new customer behavior